

INFRATHRONE (COREOPS TRACK)

WEEK 1 ASSIGNMENT PACK (Linux + Kubernetes Fundamentals)

Theme: Building the Foundation of Outage Handling

This assignment is designed to push you slightly outside your comfort zone while keeping everything achievable.

You must submit screenshots, notes, and your debugging explanation for each task.

The goal is not perfection; the goal is understanding.



K8s Networking & DNS Debugging

Objective:

Understand Service → Endpoint → Pod flow and how to debug failures.

Tasks:

1. Create a service: **kubectl expose pod web --port=80 --target-port=80 --name=web-svc**
2. Exec into a pod: **kubectl exec -it web -- bash**
3. Test:
 - a. **nslookup web-svc**
 - b. **curl web-svc**
4. Now break it intentionally: **kubectl delete pod web**
Immediately run:
 - a. **kubectl get ep web-svc**
 - b. **kubectl get events**
5. Recreate pod and confirm the service works again.

Submission:

- **DNS lookup screenshot**
- **Endpoints screenshot before & after pod deletion**
- **Write: “Why Kubernetes networking failures cause 70% of outages?”**



THE DEVOPS WAR ROOM